

## SAMOS, Greece

WMO: 167230

Lat: 37.690N

Lon: 26.912E

Elev: 19

StdP: 14.69

Time Zone: 2.00 (EUE)

Period: 94-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 36.3       | 39.0 | 19.4                          | 14.6 | 41.4 | 22.9 | 17.3 | 42.6 | 29.9                  | 52.4 | 28.3 | 51.7 | 8.9                   | 0    | 0.481 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.1                   | 96.9            | 71.3 | 94.7 | 70.6 | 92.2 | 70.0 | 77.4                | 89.1 | 74.8 | 87.0 | 72.9 | 85.6 | 13.8                 | 0    |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 73.7                            | 125.6 | 85.3 | 71.3 | 115.5 | 82.1 | 68.4 | 104.5 | 80.7 | 40.9          | 89.3 | 38.4 | 86.8 | 36.7 | 85.3 | 85.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 24.3              | 22.4 | 20.7 | DB                         | 32.5 | 102.9              | 3.0 | 3.4                                                | 30.3 | 105.3      | 28.6 | 107.3      | 27.0 | 109.1      | 24.8 | 111.6 |
|                   |      |      | WB                         | 28.9 | 77.4               | 2.8 | 1.9                                                | 26.8 | 78.8       | 25.1 | 79.9       | 23.5 | 81.0       | 21.5 | 82.4  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 65.4        | 51.1  | 51.1  | 53.5  | 59.1  | 69.0  | 78.0  | 83.4  | 83.2  | 75.4  | 67.3  | 58.9  | 53.8  | (6)  |
|                                                                        | DBStd       | 12.92       | 5.93  | 5.62  | 5.14  | 4.96  | 5.03  | 4.83  | 3.93  | 4.54  | 4.48  | 5.22  | 5.95  | 5.81  | (7)  |
|                                                                        | HDD50       | 179         | 58    | 50    | 24    | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 12    | 32    | (8)  |
|                                                                        | HDD65       | 1958        | 432   | 389   | 356   | 186   | 18    | 0     | 0     | 0     | 0     | 37    | 191   | 349   | (9)  |
|                                                                        | CDD50       | 5798        | 91    | 81    | 134   | 275   | 588   | 841   | 1035  | 1029  | 761   | 535   | 279   | 149   | (10) |
|                                                                        | CDD65       | 2102        | 0     | 0     | 1     | 8     | 141   | 391   | 570   | 564   | 311   | 108   | 8     | 0     | (11) |
|                                                                        | CDH74       | 21742       | 0     | 0     | 1     | 30    | 898   | 3999  | 7114  | 6872  | 2421  | 403   | 4     | 0     | (12) |
|                                                                        | CDH80       | 9315        | 0     | 0     | 0     | 1     | 185   | 1566  | 3487  | 3331  | 696   | 49    | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 10.2        | 10.6  | 10.8  | 10.5  | 8.1   | 8.5   | 10.5  | 11.8  | 12.4  | 9.4   | 9.5   | 9.4   | 11.4  | (14) |
| (15) Precipitation                                                     | PrecAvg     | 28.3        | 5.2   | 4.1   | 3.0   | 2.0   | 1.4   | 0.1   | 0.0   | 0.0   | 0.4   | 1.8   | 4.1   | 6.3   | (15) |
|                                                                        | PrecMax     | 40.4        | 12.6  | 9.1   | 6.5   | 4.5   | 5.1   | 0.5   | 0.3   | 0.1   | 2.4   | 5.8   | 16.8  | 15.5  | (16) |
|                                                                        | PrecMin     | 15.9        | 0.8   | 0.9   | 0.3   | 0.4   | 0.4   | 0.0   | 0.0   | 0.0   | 0.0   | 0.2   | 0.5   | 0.9   | (17) |
|                                                                        | PrecStd     | 6.1         | 2.9   | 2.4   | 1.8   | 1.2   | 1.1   | 0.1   | 0.0   | 0.0   | 0.7   | 1.4   | 3.2   | 3.2   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 64.8  | 64.6  | 71.2  | 78.7  | 87.8  | 96.9  | 101.8 | 101.0 | 93.0  | 84.5  | 73.8  | 66.7  | (19) |
|                                                                        |             | MCWB        | 57.4  | 55.7  | 58.1  | 60.8  | 64.7  | 72.0  | 71.2  | 71.1  | 67.7  | 65.6  | 62.5  | 59.4  | (20) |
|                                                                        | 2%          | DB          | 63.0  | 62.8  | 66.6  | 73.7  | 84.5  | 93.6  | 96.8  | 97.3  | 89.5  | 80.7  | 71.4  | 65.4  | (21) |
|                                                                        |             | MCWB        | 56.9  | 55.5  | 56.6  | 58.6  | 64.3  | 69.3  | 71.6  | 70.8  | 66.7  | 64.9  | 62.3  | 58.6  | (22) |
|                                                                        | 5%          | DB          | 62.2  | 61.2  | 64.4  | 71.2  | 82.0  | 91.2  | 94.7  | 94.8  | 86.4  | 77.5  | 69.7  | 64.1  | (23) |
|                                                                        |             | MCWB        | 56.4  | 55.0  | 54.9  | 58.6  | 63.8  | 68.4  | 71.4  | 70.1  | 66.6  | 64.6  | 61.6  | 58.2  | (24) |
|                                                                        | 10%         | DB          | 60.7  | 60.3  | 62.6  | 68.3  | 78.9  | 88.1  | 92.1  | 92.0  | 84.1  | 75.6  | 68.0  | 62.7  | (25) |
|                                                                        |             | MCWB        | 55.2  | 54.5  | 54.9  | 58.0  | 62.3  | 66.9  | 70.8  | 69.9  | 65.8  | 64.0  | 61.0  | 57.2  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 59.6  | 58.5  | 60.3  | 63.0  | 68.1  | 80.7  | 81.4  | 78.8  | 73.6  | 71.0  | 65.8  | 61.4  | (27) |
|                                                                        |             | MCDB        | 63.1  | 62.4  | 67.3  | 72.3  | 80.1  | 93.6  | 92.3  | 89.9  | 80.8  | 77.5  | 70.5  | 65.5  | (28) |
|                                                                        | 2%          | WB          | 57.9  | 57.0  | 58.6  | 61.2  | 66.7  | 74.5  | 78.6  | 75.8  | 72.0  | 69.2  | 64.5  | 59.7  | (29) |
|                                                                        |             | MCDB        | 62.2  | 61.1  | 64.2  | 69.4  | 78.4  | 85.4  | 90.2  | 87.9  | 79.9  | 75.7  | 69.5  | 64.0  | (30) |
|                                                                        | 5%          | WB          | 56.5  | 55.8  | 57.2  | 60.1  | 65.7  | 71.0  | 75.9  | 74.0  | 70.8  | 67.9  | 63.3  | 58.7  | (31) |
|                                                                        |             | MCDB        | 61.2  | 60.2  | 62.6  | 68.0  | 76.9  | 82.2  | 87.9  | 87.2  | 79.7  | 74.6  | 68.2  | 63.2  | (32) |
|                                                                        | 10%         | WB          | 55.1  | 54.7  | 55.8  | 58.9  | 64.6  | 69.3  | 73.6  | 72.5  | 69.3  | 66.5  | 61.7  | 57.3  | (33) |
|                                                                        |             | MCDB        | 59.9  | 59.6  | 61.4  | 66.5  | 75.4  | 83.1  | 86.6  | 86.8  | 79.5  | 73.6  | 66.9  | 62.2  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 10.4  | 11.5  | 12.7  | 14.3  | 16.8  | 17.4  | 17.1  | 16.9  | 15.7  | 13.5  | 12.2  | 9.7   | (35) |
|                                                                        |             | MCDBR       | 10.3  | 12.3  | 15.2  | 18.5  | 19.5  | 19.4  | 19.0  | 18.8  | 17.9  | 15.4  | 13.2  | 10.1  | (36) |
|                                                                        | 5% WB       | MCWBR       | 7.3   | 8.8   | 9.5   | 9.4   | 8.3   | 9.1   | 8.9   | 7.7   | 8.2   | 8.6   | 8.6   | 7.4   | (37) |
|                                                                        |             | MCWBR       | 10.2  | 11.1  | 13.4  | 15.6  | 17.6  | 18.5  | 17.7  | 17.5  | 15.9  | 13.8  | 11.8  | 8.2   | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.367 | 0.401 | 0.438 | 0.474 | 0.452 | 0.414 | 0.411 | 0.418 | 0.410 | 0.409 | 0.386 | 0.372 | (40) |
|                                                                        |             | taud        | 2.374 | 2.257 | 2.145 | 2.081 | 2.204 | 2.361 | 2.369 | 2.338 | 2.362 | 2.329 | 2.350 | 2.370 | (41) |
|                                                                        | Ebn at Noon |             | 254   | 259   | 260   | 257   | 265   | 275   | 275   | 270   | 266   | 253   | 244   | 243   | (42) |
|                                                                        |             | Edh at Noon | 29    | 37    | 45    | 51    | 46    | 39    | 38    | 39    | 36    | 34    | 30    | 27    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 685   | 940   | 1410  | 1839  | 2228  | 2521  | 2549  | 2263  | 1813  | 1285  | 858   | 627   | (44) |
|                                                                        | RadStd      |             | 74    | 82    | 89    | 152   | 107   | 100   | 39    | 38    | 79    | 63    | 64    | 85    | (45) |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |
|      |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m)  |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   | (46) |
| (47) | Regional (16 neighbors) | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -112  | N/A   | N/A   | (47) |

Nomenclature: See separate page